

FMCW Radar Based Age Group Classification System for Child Presence Detection Using Deep Learning

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Abstract—In recent years, there has been an increase in the number of cases where children have been left in cars leading to their deaths. The emergence of Child Presence Detection (CPD) systems has become an important way to improve the automotive safety, by using various sensors to detect whether a child has been left inside the car and issuing alerts or other interventions. Millimeter-wave radar can detect subtle movements inside the vehicle corresponding to human vital signs, which is widely used in the current research on CPD systems. In response to the shortcomings of current methods for detecting children left in cars, we proposed a children age group classification system based on deep learning method. In this work, we extracted the frequency domain features of raw radar signals by range fast Fourier transform (FFT) and Doppler FFT, and proposed a Multi-scale Residual Network (MSRN) to classify children of different age groups. The method proposed in this study was validated on a children vital sign dataset based on frequency-modulated continuous wave (FMCW) radar, and the average accuracy can achieve 97.43% test on a 2-class task by using range-Doppler map. The results show that the proposed age group classification system can effectively recognize children of different ages, can be applied to the development of CPD systems for intelligent vehicle, and contribute to the safety ensuring of children in cars.

Keywords—Child Presence Detection (CPD), Frequency-Modulated Continuous Wave (FMCW) Radar, Fast Fourier Transform (FFT), Multi-Scale Residual Network (MSRN), Deep Learning

I. INTRODUCTION

As an indispensable means of transportation in modern life, automobiles have greatly improved people's travel efficiency and convenience. However, with the widespread use of automobiles comes a series of problems and challenges. Among them, the heatstroke and death caused by children left alone in cars has attracted more and more attention and discussion in worldwide. According to statistics, there were as many as 141 incidents of children trapped in cars occurred in China between 2013-2018, which include children being forgotten in car due to the parental negligence, children being intentionally left in car, and in rare cases, children accidentally locking themselves in the carriage or luggage compartment. Studies have shown that the temperature inside the car would rapidly increase to 65 °C when exposure to sunlight for just ten minutes [1]. The rapid rise in temperature causes a great threat to the safety of children's lives. Since the immature thermoregulatory ability of children, it only takes

20 minutes could cause damage to the child's brain and kidneys in such high temperature environment, and 30 minutes may coma or even death. In addition, children lack sufficient self-rescue skills and are more likely to suffer life-threatening injuries compared to adults when dealing with trapped situations.

Child Presence Detection (CPD) systems have emerged as a necessary approach to improve the safety of automobiles. By installing multiple sensors in vehicle, CPD systems can monitor vital signs inside the car to detect whether a child has been left, and alert the vehicle owner by seeding message, emitting beeps, and contacting rescue organizations. In addition, some CPD systems also provide the functions like controlling the window or turning on air conditioner, which can effectively lower the temperature inside the car and improve the survival chances of trapped children. Such timely and effective intervention can reduce the risk to certain extent, and prevent the accidents of children being trapped and die in car. The CPD test items has been added to the 2024 version of China-New Car Assessment Program (C-NCAP), providing additional points for vehicles equipped with such functional devices. This policy not only encourages automobiles to innovate in safety technology, but also raises consumers' attention to child safety. With the development of modern technology, CPD system plays an important role in car safety improving and protecting children's lives, becoming an indispensable part of modern automotive safety requirement.

Early CPD systems were mainly implemented by various sensors equipped on child seats, such as optical sensors and pressure sensors [2][3]. However, this sensor-based method has the disadvantages of limited coverage and high false alarm rate. Hard to distinguish children from objects with similar shape, resulting in unsatisfactory performance of this method in practical applications. To address these issues, direct detection methods are introduced, which can directly detect human vital signs like heartbeat, breathing, or movement, which can effectively reduce the false alarm rate of the system. Infrared sensor-based methods [4][5] have been proven to extend the coverage by detecting children's movements. However, infrared sensors are susceptible to interference from surrounding heat and light sources, especially when the temperature of environment is close to the human body, the detection accuracy will significantly decrease. Vision-based CPD systems provide more reliable

performance by using image processing method. However, it often suffers from privacy protection issues. In addition, there are physical defects when the camera is obstructed by the line of sight. Recently, millimeter-wave radar-based human sensing technology in vehicle has been widely explored in vital signs detection and position estimation. Millimeter-wave radar can detect subtle movements corresponding to human vital signs like respiration and heartbeat, which can not only identify whether there is a child left in the car, but also predict its location and vital signs. Meanwhile, millimeter-wave radar-based CPD systems have the advantages in terms of privacy protection and no need to wear sensors. After the vital signs have been successfully extracted by millimeter-wave radar, which can be further analyzed by deep learning methods. Ahmed *et al.* used deep learning classifiers to classify seven activities performed at different distances from the radar. The received radar data was fed into a feature extraction block with three convolutional layers and softmax was performed, which resulted in 91% recognition accuracy.

In this paper, we proposed a deep learning-based age group classification system for children left in car in response to the development of CPD system, and which is validated on a children vital sign dataset based on millimeter-wave radar. We used range FFT and Doppler FFT to extract the frequency domain information of the radar signals, and the range-Doppler map are inputted into the Multi-scale Residual Network (MSRN) for classification to detect children of different age groups[6].

This paper is organized as follows. Section II describes the dataset used in this paper and introduces the method of feature extraction and classification. The results of children age group classification are presented in Section III. Section IV gives a brief conclusion.

II. METHOD

A. Dataset

In this paper, we use the Children Vital Sign public dataset provided by Yoo *et al.* This dataset contains the vital sign data of 50 children under 13 years old recorded by millimeter-wave radar. The 50 subjects include 24 boys and 26 girls, with a mean age of 5 years old and a mean body mass index (BMI) of 16.68. The radar system used in this dataset is equipped with a frequency-modulated continuous wave (FMCW) radar sensor, IWR-6843, designed and manufactured by Texas Instrument (TI), USA, and use one transmitter and four receivers, with 60.25 GHz of starting frequency and 20 frames per second (FPS). During the experiments, each subject was sitting in a chair, or a child seat for the subject under the age of 6.

B. FMCW Radar Signal Processing

FMCW radar is a radar system that detect targets and measure distances by transmitting and receiving linear frequency modulated signals, which transmits $x_T(t)$, a continuous-wave signal with a linearly increasing frequency, commonly referred to as chirp signal. The chirp's frequency starts from an initial rate, and increases at a constant rate, covering a bandwidth, the transmitted signal can be expressed as:

$$x_T(t) = A_T \cos(2\pi f_c(t) + \pi \frac{B}{T_c} t^2 + \phi) \quad (1)$$

Where A_T denotes the transmitted power of the signal, f_c denotes the initial frequency, B denotes the bandwidth, and T_c is the duration of the chirp.

The transmitted chirp is reflected back when it encounters the target and be received by the radar's receiver. When the signal reflected by the target within the operating range of radar, the reflected signal $x_R(t)$ can be expressed as:

$$x_R(t) = \alpha A_T \cos(2\pi f_c(t - t_d) + \pi \frac{B}{T_c} (t - t_d)^2 + \phi) \quad (2)$$

Where α and t_d denote the attenuation and time delay during round-trip of the signal, respectively.

The FMCW radar uses an analog mixer to multiply the received signal with the transmitted signal, and processes it with a low-pass filter to obtain a product signal containing the target information called the intermediate frequency (IF) signal. The frequency of IF signal is proportional to the distance to the target, which can be calculated by analyzing the frequency of the IF signal. The IF signal $x_{IF}(t)$ can be expressed as follows:

$$x_{IF}(t) = A_{IF} e^{f(2\pi \frac{B}{T_c} t_d + 2\pi f_c t_d)} \quad (3)$$

Where A_{IF} denotes the signal power of $x_{IF}(t)$.

The point to detect targets for FMCW radars is the processing of IF signal, achieved by performing Fast Fourier Transform (FFT) on the sample dimensions of the raw radar data, known as range FFT. The transmitted chirp signal is reflected upon encountering the target, and the received signal is mixed with the transmitted signal to get the IF signal. By range FFT, the distance information of the IF signal can be expressed in the form of frequency spectrum. The peak frequency of the spectrum is proportional to the target distance, and the distance to the target can be determined by analyzing the range-time map.

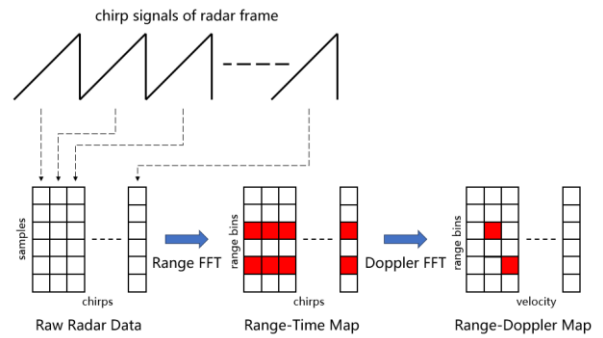


Fig.1 Processing of range FFT and Doppler FFT on chirp signals

Beside the distance, FMCW radar can also measure the relative velocity by analyzing a series of chirps within one frame. In the presence of relative motion of a target, the phase and frequency of the reflected signal experience shifts due to the Doppler effect, which results from the difference in the distance over the signal transmission. To obtain this frequency variation, Doppler FFT always be performed after range FFT, applied to multiple chirps to detect the frequency variations caused by the target's movement. By combining the results of the range FFT and Doppler FFT, the distance and relative velocity of the target can be detected at the same time.

The processing of raw radar data is shown in Figure 1. The received chirps are combined to a matrix, where the rows and columns represent the number of samples in a chirp and the number of chirps, respectively. Range FFT is a one-dimensional FFT performed on the sample dimensions of the data matrix to extract the distance information of the target. After accumulating all the frames, range-time map can be obtained. The peak values in the map correspond to the distance between the target and the radar. After performing the range FFT, the relative velocity can be extracted by applying Doppler FFT for each range bin to obtain the range-Doppler map, providing a comprehensive view of both the target's distance and velocity. Finally, the range-Doppler map is used as the input to the deep learning model for age group classification.

C. Classification

Convolutional Neural Networks (CNN) as a widely used deep learning model, has shown excellent performance and wide application prospects in various fields, like computer vision, object detection, and natural language processing. CNN are typically composed by multiple hidden layers with different functions, including convolutional layer, pooling layer, and fully connected layer. These hidden layers work together to extract and process the feature of input data, achieving the automation of complex tasks.

Convolutional layer, serve as the foundation of CNN, can extract local feature of input data through convolutional operations. Each convolutional filter slides over the input with different kernels, to generate feature maps. The local receptive fields and parameter sharing mechanisms make convolutional layers efficient in capturing spatial dimensions in the data, while reducing the number of parameters needed in network and improves the computational efficiency. The convolutional operation can be expressed as:

$$y_{i,j}^l = \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} \omega_{n,m}^l \times x_{i+n,j+m}^{l-1} + b \quad (4)$$

Where $x_{i+n,j+m}^{l-1}$ denotes the element at $(i+n, j+m)$ of layer $l-1$, $y_{i,j}^l$ denotes the element at (i,j) of layer l , ω denotes the convolution kernel of size $N * M$, $\omega_{n,m}$ denotes the weight at (n,m) of convolution kernel, b is the offset.

The feature maps generated by convolutional layers pass through activation functions, typically ReLU, Sigmoid, or Tanh, introducing non-linearity to the network and allowing it to learn complex patterns. Following the convolutional layers, pooling layers are often applied to down-sample the feature maps, reducing spatial dimensions while retaining the important features, including Max Pooling and Average Pooling. The pooling process can not only reduce the computational load but also provide a degree of translation invariance. Eventually, the outputs of convolutional layers and pooling layers are passed by fully connected layers, serve as the classifier of the network, which aggregate the flattened features learned by convolutional layers and mapping them to class probabilities or regression outputs depending on the task.

In deep neural networks, increasing layers theoretically should improve the performance. However, as the network deepens, they can encounter the performance of the model decrease rather than improve, known as the degradation problem. Residual blocks are introduced to address this issue

by introducing identity mapping and skip connection. The core idea of residual block is to allow the network to learn residuals between the input and output, instead of directly mapping from the input to output. In this way, the gradient can propagate stably in the network, mitigating issues like vanishing or exploding gradients, while enhance the model performance.

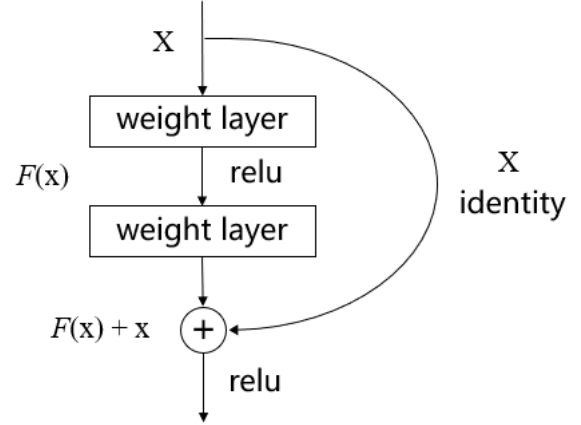


Fig.2 The diagram of residual block

The diagram of residual block is shown in Figure 2. In the residual block, the input is directly added to the output of a series of convolutional layers, effectively letting the network learn the residual function, denoted as $F(x)$, where the original mapping is represented as $H(x) = F(x) + x$. This approach makes it easier for the network to learn since it focuses on the difference between the input and the desired output, rather than trying to learn a complex mapping. When the network is already optimized, setting the residual to zero ensures that the identity mapping is preserved, preventing performance degradation and maintaining at least the current level of performance.

The range-time map and range-Doppler map obtained by range FFT and Doppler FFT are put into the Multi-scale Residual Network (MSRN) for classification. The proposed MSRN architecture is illustrated in Figure 3. The input data passes through a Convolution Block (Conv Block) for feature extraction, including a convolutional layer with the kernel width of 7, a Batch Normalization (BN) layer to prevent overfitting, and ReLU activation to enhance non-linearity of the model. After the Conv Block, a max-pooling layer is applied for dimensionality reduction. The down-sampled feature matrix is then inputted to the Multi-scale Residual Block (MSR Block), comprised by multiple Conv Blocks with different kernel sizes to extract features at various scales. The outputs of different paths are concatenated along the channel dimension, integrating information from multiple receptive fields. Subsequently, a convolutional layer is used to map the concatenated result back to the same dimensionality as the input, which is added to the original input via residual connections. The MSR Block can improve the model's ability to capture features in different spatial dimensions and prevent degradation problem at the same time. At the end of the model, a Fully Connected (FC) layer is used to map the high-dimensional feature matrix to the output space and softmax function is applied to obtain the probability distribution between different age group, achieving the final decision of the classification task.

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} J(\theta; x^{(i)}, y^{(i)}) \quad (5)$$

Where θ_t represents the model parameters at iteration t , η is the learning rate, and $\nabla_{\theta} J(\theta; x^{(i)}, y^{(i)})$ is the gradient of the loss function with respect to the model parameters for the selected training sample $(x^{(i)}, y^{(i)})$.

III. RESULTS

The children vital sign dataset contained 50 children under 13 years old, and three different age group classifications are performed. In the first experiment, the children in the vital sign dataset are divided into four groups: under 3 years, 3 to 6 years, 6 to 9 years, and 9 to 13 years. In the second experiment, the children are divided into three groups: under 4 years, 4 to 8 years, and 8 to 13 years. Finally, in the third experiment, two age groups are established: 0 to 6 years and 6 to 13 years.

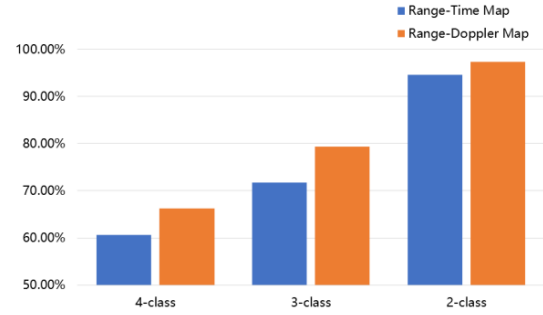


Fig.4 The classification results by using range-time map and range-Doppler map

In the study that provided the dataset, Yoo *et al.* used GoogLeNet to classify the above three sets of experiments. Besides, Nguyen *et al.* extracted 39 features on time-domain and frequency-domain from the raw radar signals, which form a high-dimensional feature vector, and used Random Forest classifier trained based on these features for the classification of children age groups in the dataset, as shown in Table I.

TABLE I. THE PERFORMANCE COMPARISON WITH OTHER METHOD

Method	4 class	3 class	2 class
GoogLeNet	58.50%	68.00%	96.25%
Random Forest	40.04%	-	-
MSRN	66.29%	79.43%	97.43%

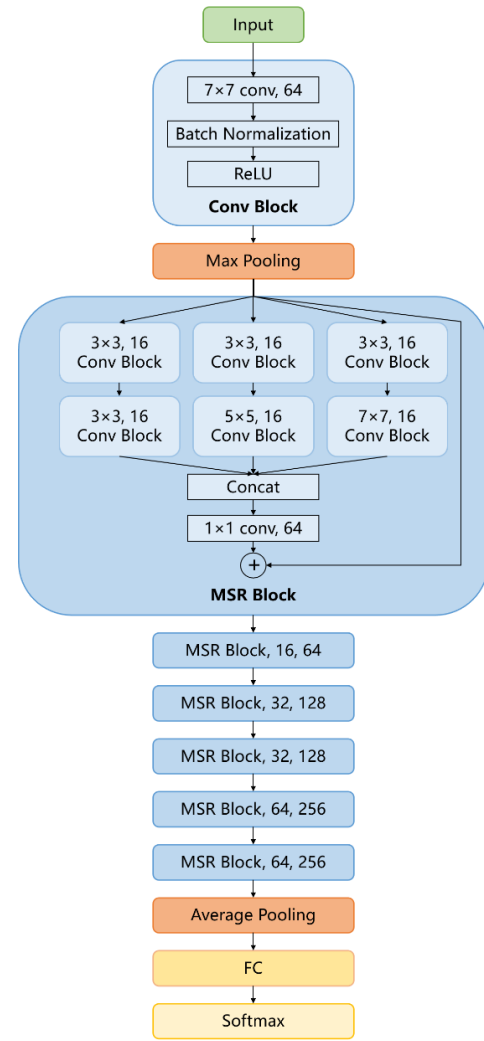


Fig.3 The architecture of proposed MSRN

In this study, cross-entropy is chosen as the loss function, and the Stochastic Gradient Descent (SGD) algorithm is used to optimize the network parameters. The SGD algorithm estimates the gradient by using randomly selected training samples in each iteration, updating the model parameters to minimize the loss function. Compared to the standard gradient descent algorithm, SGD can accelerate convergence, and prevent falling into local optima at the same time. The SGD algorithm can be expressed as:

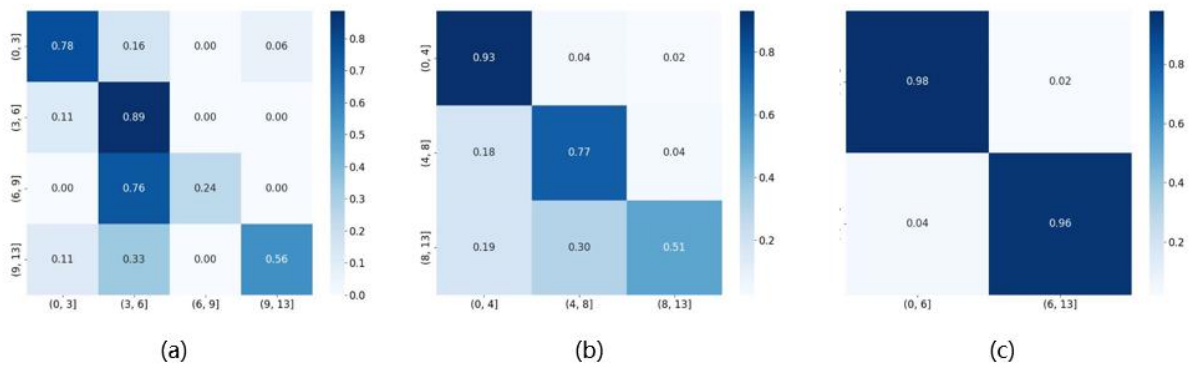


Fig.5 The confusion matrices of different sets of age groups. (a) 4-class. (b) 3-class. (c) 2-class.

In addition, we compare the effect of frequency domain features extracted by two different FFT method on the results

of children age group classification. The range-time map obtained by range FFT and the range-Doppler map obtained by Doppler FFT are used as inputs to the MSRN for classification. The classification results are shown in Figure 4. It can be seen that the classification performance of using range-Doppler map is consistently better than range-time map. The results show that when using FMCW radar for the detection of children in vehicles, the velocity information extracted by Doppler FFT is better compared to the range-time map that contains only range information. By performing Doppler FFT on the range-time map obtained to obtain velocity information, the classification accuracy of children age detection can be effectively improved.

The confusion matrices of three age group sets of experiments using range-Doppler map are shown in Figure 5. As the number of age groups decreased, the classification performance increased significantly. In the confusion matrix of 2-class task, the accuracy of children aged 0-6 years as well as the above 6 years aged can exceed 96%, with only a small rate of confusion between the samples. The results prove that the model shows great performance in the prediction of both positive and negative samples, and the proposed children age group classification system can effectively detect a child is whether older than 6 years old or not.

IV. CONCLUSIONS

To solve the problem that current automotive CPD system has insufficient protection measures, which leads to a large number of children being left in car and killed, this paper proposed a deep learning-based age group classification system for children left in car. Range FFT and Doppler FFT are performed on the collected FMCW radar signals of 50

children to extract the frequency domain features, and a deep learning model based on Multi-scale Residual Network is proposed to classify the range-time map and range-Doppler map. The proposed MSRN model demonstrates a high level of accuracy over using other method, particularly when using range-Doppler maps for classification, achieving an average accuracy of 97.43%. The results indicate that proposed age group classification system can effectively classify children across different age groups, enabling timely and appropriate interventions to prevent tragic incidents of children being left unattended in vehicles. The proposed method shows significant potential for integration into future CPD systems, and contributing to enhanced child safety in automotive environments.

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